

AI Requirement Engineering Report

BUSINESS ANALYSIS

****Business Analysis Report: Online Food Delivery Application****

****Project Overview****

The online food delivery application project aims to design, develop, and launch a user-friendly platform that enables customers to order food from various restaurants and have it delivered at their doorstep. The platform will provide a seamless experience for both customers and restaurants, increasing the reach and revenue of participating restaurants while providing customers with a convenient and efficient way to enjoy their favorite food.

****Problem Statement****

The current food delivery landscape is dominated by traditional methods, such as phone orders and cash payments, which can be time-consuming and inconvenient. The lack of a unified platform has led to fragmented experiences for both customers and restaurants, resulting in missed opportunities for growth and revenue. The online food delivery application project seeks to address these issues by providing a comprehensive solution that integrates multiple stakeholders and streamlines the ordering process.

****Business Objectives****

1. ****Increase Customer Engagement****: To attract and retain a large customer base, increasing the frequency and average order value of customers.
2. ****Expand Restaurant Reach****: To onboard a diverse range of restaurants, increasing the number of participating restaurants and their exposure to potential customers.
3. ****Improve Order Accuracy****: To minimize errors in order processing, ensuring that customers receive their ordered food correctly and on time.
4. ****Enhance User Experience****: To design an intuitive and user-friendly interface, making it easy for customers to navigate and place orders.
5. ****Increase Revenue****: To generate significant revenue through commission-based sales and advertising, while maintaining a competitive pricing strategy.
6. ****Reduce Delivery Time****: To minimize delivery times, ensuring that customers receive their food within a reasonable timeframe.
7. ****Implement Secure Payment Processing****: To ensure secure payment processing, protecting customer data and preventing unauthorized transactions.
8. ****Monitor and Analyze Performance****: To track key performance indicators (KPIs), such as order volume, customer satisfaction, and revenue growth.

****Stakeholders****

1. ****Customers****: Individuals who will use the online food delivery application to order food from participating restaurants.
2. ****Restaurants****: Establishments that will participate in the online food delivery application, providing their menu items and pricing.
3. ****Developers****: Technical experts responsible for designing, developing, and maintaining the online food delivery application.
4. ****Marketing Team****: Professionals responsible for promoting the online food delivery application to attract customers and restaurants.
5. ****Operations Team****: Staff responsible for managing the logistics of food delivery, including inventory management and customer support.

****Stakeholder Responsibilities****

1. ****Customers****: Provide accurate order information, pay for orders in a timely manner, and provide feedback on their experience.
2. ****Restaurants****: Provide menu items, pricing, and inventory information, as well as participate in marketing and promotional activities.
3. ****Developers****: Design, develop, and maintain the online food delivery application, ensuring it meets stakeholder requirements.
4. ****Marketing Team****: Promote the online food delivery application through various channels, including social media, advertising, and partnerships.
5. ****Operations Team****: Manage the logistics of food delivery, including inventory management, customer support, and delivery scheduling.

****Business Risks****

1. ****Competition from Established Players****: The online food delivery market is highly competitive, with established players that may pose a threat to the project's success.
2. ****Technical Issues****: Technical problems, such as platform downtime or data breaches, can impact the user experience and reputation of the online food delivery application.
3. ****Restaurant Non-Participation****: Restaurants may decline to participate in the online food delivery application, reducing the platform's overall revenue potential.
4. ****Customer Disengagement****: Customers may lose interest in using the online food delivery application, reducing order volume and revenue.
5. ****Regulatory Compliance****: The online food delivery application must comply with relevant regulations, such as food safety and security standards.

****Risk Mitigation Strategies****

1. ****Conduct Market Research****: Gather data on the competitive landscape, customer preferences, and market trends to inform business decisions.
2. ****Develop a Robust Technical Infrastructure****: Invest in robust technical infrastructure, including scalable servers and reliable payment processing systems.
3. ****Establish Partnerships with Restaurants****: Foster relationships with restaurants to ensure their participation and support.
4. ****Implement Customer Retention Strategies****: Develop strategies to retain customers, such as loyalty programs and personalized marketing campaigns.
5. ****Monitor Regulatory Compliance****: Regularly review and update the online food delivery application to ensure compliance with relevant regulations.

****Assumptions****

1. ****Customers will use the online food delivery application regularly****.
2. ****Restaurants will participate in the online food delivery application****.
3. ****The online food delivery application will be developed using a scalable and reliable technical infrastructure****.
4. ****Marketing efforts will be effective in attracting customers and restaurants****.
5. ****Regulatory compliance will be maintained throughout the project lifecycle****.

****Constraints****

1. ****Budget constraints****: The project has a limited budget, which may impact the scope and quality of the online food delivery application.
2. ****Technical limitations****: The project is subject to technical limitations, such as platform downtime or data breaches.
3. ****Restaurant participation****: The project relies on restaurant participation, which may be affected

by factors such as competition or regulatory changes.

4. **Customer engagement**: The project aims to increase customer engagement, but this may be challenging due to factors such as competition or market saturation.

5. **Regulatory compliance**: The project must comply with relevant regulations, which may require additional resources and time.

Project Scope

The online food delivery application project includes the following scope:

1. **Development of the online food delivery platform**: Design, development, and testing of the platform.
2. **Integration with payment processing systems**: Integration with secure payment processing systems.
3. **Implementation of customer retention strategies**: Development of strategies to retain customers.
4. **Marketing and promotion**: Promotion of the online food delivery application through various channels.

Out of Scope Items

1. **Development of a mobile app**: The project does not include the development of a mobile app.
2. **Integration with third-party services**: The project does not include integration with third-party services, such as social media or review platforms.
3. **Development of a loyalty program**: The project does not include the development of a loyalty program.
4. **Implementation of a customer support system**: The project does not include implementation of a customer support system.

Expected Benefits

1. **Increased customer engagement**: The online food delivery application is expected to increase customer engagement, leading to increased order volume and revenue.
2. **Expanded restaurant reach**: The online food delivery application is expected to expand the reach of participating restaurants, increasing their exposure to potential customers.
3. **Improved user experience**: The online food delivery application is expected to provide an intuitive and user-friendly interface, making it easy for customers to navigate and place orders.
4. **Increased revenue**: The online food delivery application is expected to generate significant revenue through commission-based sales and advertising.
5. **Enhanced brand reputation**: The online food delivery application is expected to enhance the brand reputation of participating restaurants and the project team.

REQUIREMENTS

Project: Online Food Delivery Application

Project Purpose

The online food delivery application aims to provide a convenient and efficient platform for customers to order food from their preferred restaurants, while ensuring a seamless experience for restaurants and delivery partners.

Project Scope

The scope of this project includes the development of an online food delivery application that connects customers, restaurants, and delivery partners. The application will enable customers to browse menus, place orders, and track their orders in real-time. Restaurants will be able to manage their menus, track orders, and receive payment notifications. Delivery partners will be able to

manage their deliveries and track their progress.

****Functional Requirements:****

1. ****Customer Registration and Login****: The application should allow customers to register and log in to their accounts, enabling them to access their order history, favorite restaurants, and personalized recommendations.
2. ****Restaurant Listing****: The application should display a list of available restaurants, including their menus, ratings, and reviews.
3. ****Menu Management****: Restaurants should be able to manage their menus, including adding, editing, and deleting menu items.
4. ****Food Search****: Customers should be able to search for specific dishes or cuisines, with the option to filter by price, rating, and delivery time.
5. ****Order Placement****: Customers should be able to place orders online, selecting their preferred restaurants, dishes, and delivery options.
6. ****Online Payment****: The application should integrate a secure payment gateway, allowing customers to pay for their orders online.
7. ****Real-time Order Tracking****: Customers should be able to track the status of their orders in real-time, including preparation, delivery, and arrival.
8. ****Delivery Partner Management****: Delivery partners should be able to manage their deliveries, including tracking their progress, updating order status, and receiving payment notifications.
9. ****Rating and Review System****: Customers should be able to rate and review their dining experiences, providing feedback to restaurants and helping others make informed decisions.
10. ****Push Notifications****: The application should send push notifications to customers and delivery partners, keeping them informed about order updates, promotions, and special offers.

****Non-Functional Requirements:****

1. ****Real-time Updates****: The application should provide real-time updates on order status, menu changes, and promotions to ensure customers have the most up-to-date information.
2. ****High Availability****: The application should be available 24/7, with minimal downtime or errors, to ensure a seamless experience for customers and restaurants.
3. ****Secure Payments****: The application should integrate a secure payment gateway, protecting customers' financial information and ensuring a safe transaction process.
4. ****Fast Response Time****: The application should respond quickly to customer inquiries, order updates, and delivery requests, minimizing wait times and improving overall user experience.

****User Stories:****

1. ****As a customer, I want to be able to register and log in to my account so that I can access my order history and favorite restaurants.****
2. ****As a restaurant owner, I want to be able to manage my menu so that I can update my dishes and prices in real-time.****
3. ****As a delivery partner, I want to be able to track my deliveries so that I can ensure timely arrival and provide excellent customer service.****
4. ****As a customer, I want to be able to search for specific dishes or cuisines so that I can find my favorite food quickly and easily.****
5. ****As a restaurant owner, I want to be able to receive payment notifications so that I can track my orders and manage my finances effectively.****

****Business Rules:****

1. ****Minimum Order Value****: The application should enforce a minimum order value to prevent customers from placing small orders or exploiting the system.
2. ****Delivery Time Windows****: The application should allow restaurants to set delivery time windows, ensuring that customers receive their orders within a reasonable timeframe.
3. ****Rating Requirements****: The application should require customers to rate and review their

dining experiences, providing feedback to restaurants and helping others make informed decisions.

****Assumptions:****

1. ****Customer Demographics****: The application assumes that the target customer demographic is urban, tech-savvy, and active on social media.
2. ****Restaurant Availability****: The application assumes that restaurants will be available to participate in the platform, providing menus and managing orders.
3. ****Delivery Partner Availability****: The application assumes that delivery partners will be available to participate in the platform, tracking deliveries and providing excellent customer service.

****Constraints:****

1. ****Technical Limitations****: The application is constrained by technical limitations, including database storage capacity, server processing power, and network bandwidth.
2. ****Regulatory Requirements****: The application is subject to regulatory requirements, including data protection laws and payment processing regulations.
3. ****Scalability****: The application should be designed to scale with increasing demand, ensuring that it can handle a large number of customers, restaurants, and delivery partners.

SRS

1. Introduction

* Purpose: The purpose of this document is to provide a comprehensive Software Requirement Specification (SRS) for the Online Food Delivery Application project.

* Scope: The scope of this project includes the development of an online food delivery application that connects customers, restaurants, and delivery partners.

* Definitions:

- + Customer: An individual who places orders through the application.
- + Restaurant: A business that provides food services to customers through the application.
- + Delivery Partner: An individual or entity responsible for delivering food to customers.

2. Overall Description

* Product Perspective: The online food delivery application is a software product designed to facilitate food ordering and delivery between customers, restaurants, and delivery partners.

* Product Functions:

- + Customer registration and login
- + Restaurant listing and menu management
- + Food search and ordering
- + Online payment processing
- + Real-time order tracking
- + Delivery partner management

* User Classes:

- + Customers
- + Restaurants
- + Delivery Partners

* Operating Environment:

- + Web-based application with mobile app integration
- + Database storage for customer, restaurant, and delivery partner data
- + Payment gateway integration

3. Functional Requirements

FR-1: Customer Registration and Login

- * The application should allow customers to register and log in to their accounts.
- * The application should provide a secure login process to protect customer data.

FR-2: Restaurant Listing

- * The application should display a list of available restaurants, including their menus, ratings, and reviews.
- * The application should allow restaurants to update their menu information and ratings.

FR-3: Menu Management

- * Restaurants should be able to manage their menus, including adding, editing, and deleting menu items.
- * The application should provide a user-friendly interface for restaurant menu management.

FR-4: Food Search

- * Customers should be able to search for specific dishes or cuisines.
- * The application should filter search results by price, rating, and delivery time.

FR-5: Order Placement

- * Customers should be able to place orders online, selecting their preferred restaurants, dishes, and delivery options.
- * The application should provide a secure payment processing system.

FR-6: Real-time Order Tracking

- * Customers should be able to track the status of their orders in real-time.
- * The application should provide updates on order preparation, delivery, and arrival.

FR-7: Delivery Partner Management

- * Delivery partners should be able to manage their deliveries, including tracking their progress and updating order status.
- * The application should provide payment notifications to delivery partners.

FR-8: Rating and Review System

- * Customers should be able to rate and review their dining experiences.
- * The application should provide a rating system and allow customers to leave reviews.

FR-9: Push Notifications

- * The application should send push notifications to customers and delivery partners, keeping them informed about order updates, promotions, and special offers.

4. Non Functional Requirements

NFR-1: Real-time Updates

- * The application should provide real-time updates on order status, menu changes, and promotions.
- * The application should minimize latency and ensure fast response times.

NFR-2: High Availability

- * The application should be available 24/7, with minimal downtime or errors.
- * The application should provide a high level of reliability and uptime.

NFR-3: Secure Payments

- * The application should integrate a secure payment gateway to protect customer financial information.
- * The application should ensure a safe and secure transaction process.

NFR-4: Fast Response Time

- * The application should respond quickly to customer inquiries, order updates, and delivery requests.
- * The application should minimize wait times and improve overall user experience.

5. External Interface Requirements

- * User Interface:

- + Web-based application with mobile app integration
- + Intuitive and user-friendly interface for customers, restaurants, and delivery partners
- * Software Interface:
 - + API integration for payment gateway and database storage
 - + Secure communication protocols to protect data transmission
- * Hardware Interface:
 - + Compatible with a range of devices and operating systems
 - + Optimized for performance and reliability
- * Communication Interface:
 - + Integration with third-party services, including payment gateways and delivery partners

6. Security Requirements

- * Data Protection: The application should protect customer and restaurant data from unauthorized access and breaches.
- * Authentication and Authorization: The application should provide secure login and authentication processes to protect user accounts.
- * Payment Security: The application should integrate a secure payment gateway to protect customer financial information.

7. Performance Requirements

- * Response Time: The application should respond quickly to customer inquiries, order updates, and delivery requests.
- * Throughput: The application should handle a high volume of orders and transactions without significant delays or errors.
- * Scalability: The application should be designed to scale with increasing demand, ensuring that it can handle a large number of customers, restaurants, and delivery partners.

8. Assumptions

- * Customer Demographics: The application assumes that the target customer demographic is urban, tech-savvy, and active on social media.
- * Restaurant Availability: The application assumes that restaurants will be available to participate in the platform, providing menus and managing orders.
- * Delivery Partner Availability: The application assumes that delivery partners will be available to participate in the platform, tracking deliveries and providing excellent customer service.

9. Constraints

- * Technical Limitations: The application is constrained by technical limitations, including database storage capacity, server processing power, and network bandwidth.
- * Regulatory Requirements: The application is subject to regulatory requirements, including data protection laws and payment processing regulations.
- * Scalability: The application should be designed to scale with increasing demand, ensuring that it can handle a large number of customers, restaurants, and delivery partners.

10. Future Enhancements

- * Integration with new payment gateways and services
- * Expansion to new markets and regions
- * Development of additional features and functionality, including loyalty programs and personalized recommendations

ARCHITECTURE

Software Architecture

The online food delivery application will be designed using a microservices architecture, with a focus on scalability, security, and maintainability.

****High Level Architecture****

The high-level architecture consists of the following components:

1. ****Customer Portal****: A web-based interface for customers to browse menus, place orders, and track their orders.
2. ****Restaurant Portal****: A web-based interface for restaurants to manage their menus, track orders, and receive payment notifications.
3. ****Delivery Partner Portal****: A web-based interface for delivery partners to manage their deliveries and track their progress.
4. ****Order Management System****: A centralized system for managing orders, including order placement, tracking, and fulfillment.
5. ****Payment Gateway****: A secure payment gateway for processing online payments.
6. ****Database****: A relational database for storing customer, restaurant, and delivery partner data.

****System Components****

The system components include:

1. ****Customer Service API****: An API for handling customer-related requests, such as order placement and tracking.
2. ****Restaurant Service API****: An API for handling restaurant-related requests, such as menu management and order tracking.
3. ****Delivery Partner Service API****: An API for handling delivery partner-related requests, such as delivery tracking and payment notifications.
4. ****Order Management Service****: A service for managing orders, including order placement, tracking, and fulfillment.
5. ****Payment Gateway Service****: A service for processing online payments.

****Frontend Architecture****

The frontend architecture consists of the following components:

1. ****Customer Portal****: A web-based interface for customers to browse menus, place orders, and track their orders.
2. ****Restaurant Portal****: A web-based interface for restaurants to manage their menus, track orders, and receive payment notifications.
3. ****Delivery Partner Portal****: A web-based interface for delivery partners to manage their deliveries and track their progress.

****Backend Architecture****

The backend architecture consists of the following components:

1. ****Customer Service API****: An API for handling customer-related requests, such as order placement and tracking.
2. ****Restaurant Service API****: An API for handling restaurant-related requests, such as menu management and order tracking.
3. ****Delivery Partner Service API****: An API for handling delivery partner-related requests, such as delivery tracking and payment notifications.
4. ****Order Management Service****: A service for managing orders, including order placement, tracking, and fulfillment.
5. ****Payment Gateway Service****: A service for processing online payments.

****Database Architecture****

The database architecture consists of the following components:

1. ****Customer Table****: A table for storing customer data, including contact information and order history.
2. ****Restaurant Table****: A table for storing restaurant data, including menu items and contact information.
3. ****Delivery Partner Table****: A table for storing delivery partner data, including contact information and delivery history.
4. ****Order Table****: A table for storing order data, including order status and payment information.

****API Architecture****

The API architecture consists of the following components:

1. ****Customer Service API****: An API for handling customer-related requests, such as order placement and tracking.
2. ****Restaurant Service API****: An API for handling restaurant-related requests, such as menu management and order tracking.
3. ****Delivery Partner Service API****: An API for handling delivery partner-related requests, such as delivery tracking and payment notifications.
4. ****Order Management Service****: A service for managing orders, including order placement, tracking, and fulfillment.

****Authentication Flow****

The authentication flow consists of the following steps:

1. ****Customer Registration****: Customers register for an account by providing contact information and payment details.
2. ****Restaurant Registration****: Restaurants register for an account by providing contact information and menu data.
3. ****Delivery Partner Registration****: Delivery partners register for an account by providing contact information and delivery history.
4. ****Login****: Customers, restaurants, and delivery partners log in to their accounts using their registered credentials.

****User Workflow****

The user workflow consists of the following steps:

1. ****Customer Places Order****: Customers browse menus, select dishes, and place orders online.
2. ****Restaurant Receives Order****: Restaurants receive order notifications and manage their orders accordingly.
3. ****Delivery Partner Picks Up Order****: Delivery partners pick up orders and track their progress.
4. ****Customer Tracks Order****: Customers track the status of their orders in real-time.

****Deployment Architecture****

The deployment architecture consists of the following components:

1. ****Cloud Hosting****: The application will be hosted on a cloud platform, such as AWS or Azure.
2. ****Containerization****: The application will be containerized using Docker.
3. ****Orchestration****: The application will be orchestrated using Kubernetes.

****Technology Stack****

The technology stack consists of the following components:

- 1. **Programming Language**: Java or Python.
- 2. **Framework**: Spring Boot or Django.
- 3. **Database**: MySQL or PostgreSQL.
- 4. **API Framework**: Spring Boot or Django Rest Framework.
- 5. **Frontend Framework**: React or Angular.

Database Tables

The database tables consist of the following components:

- 1. **Customer Table**: A table for storing customer data, including contact information and order history.
- 2. **Restaurant Table**: A table for storing restaurant data, including menu items and contact information.
- 3. **Delivery Partner Table**: A table for storing delivery partner data, including contact information and delivery history.
- 4. **Order Table**: A table for storing order data, including order status and payment information.

Security Architecture

The security architecture consists of the following components:

- 1. **Authentication**: The application will use a secure authentication mechanism, such as OAuth or JWT.
- 2. **Authorization**: The application will use a secure authorization mechanism, such as role-based access control (RBAC).
- 3. **Data Encryption**: The application will use data encryption to protect sensitive data, such as payment information.

Scalability Considerations

The scalability considerations consist of the following components:

- 1. **Horizontal Scaling**: The application will be designed to scale horizontally, using load balancers and container orchestration.
- 2. **Vertical Scaling**: The application will be designed to scale vertically, using cloud hosting and serverless computing.
- 3. **Caching**: The application will use caching mechanisms, such as Redis or Memcached, to improve performance.

This software architecture provides a solid foundation for the online food delivery application, with a focus on scalability, security, and maintainability.

TEST CASES

Based on the provided requirements, I will generate the detailed test cases for the Online Food Delivery Application project.

Functional Test Cases

Test Case ID	Test Type	Scenario	Preconditions	Test Steps	Expected Result
FT-01	Functional	Customer Registration	Customer has no account, valid email and password.	1. The customer clicks on the "Register" button. 2. The system prompts for email and password. 3.	

The customer enters valid email and password. 4. The system generates a new account. | The system displays a success message and logs the customer in. |

| FT-02 | Functional | Order Placement | Customer has an active account, valid payment method. | 1. The customer clicks on the "Order" button. 2. The system displays the menu options. 3. The customer selects dishes and adds them to the cart. 4. The customer proceeds to checkout. 5. The system prompts for payment method. 6. The customer enters a valid payment method. 7. The system processes the payment. | The system displays a success message and updates the order status. |

| FT-03 | Functional | Menu Management | Restaurant has an active account, valid menu items. | 1. The restaurant clicks on the "Menu" button. 2. The system displays the menu options. 3. The restaurant adds, edits, or deletes menu items. 4. The system updates the menu. | The system displays a success message and updates the menu. |

| FT-04 | Functional | Food Search | Customer has an active account, valid search criteria. | 1. The customer clicks on the "Search" button. 2. The system prompts for search criteria. 3. The customer enters valid search criteria. 4. The system displays search results. | The system displays relevant search results. |

| FT-05 | Functional | Real-time Order Tracking | Customer has an active account, valid order ID. | 1. The customer clicks on the "Track Order" button. 2. The system displays the order status. 3. The customer views the order status in real-time. | The system displays the latest order status. |

****Integration Test Cases****

| Test Case ID | Test Type | Scenario | Preconditions | Test Steps | Expected Result |

|-----|-----|-----|-----|-----|-----|

| IT-01 | Integration | Payment Processing | Customer has an active account, valid payment method. | 1. The customer places an order and selects a payment method. 2. The system processes the payment. 3. The system updates the order status. | The system displays a success message and updates the order status. |

| IT-02 | Integration | Menu Management and Order Placement | Restaurant has an active account, valid menu items. | 1. The restaurant adds new menu items. 2. The customer places an order using the new menu items. 3. The system processes the payment. 4. The system updates the order status. | The system displays a success message and updates the order status. |

| IT-03 | Integration | Delivery Partner Management | Delivery partner has an active account, valid delivery location. | 1. The delivery partner receives a new delivery request. 2. The system updates the delivery partner's location. 3. The system notifies the customer of the delivery update. | The system displays a success message and updates the delivery partner's location. |

****Acceptance Test Cases****

| Test Case ID | Test Type | Scenario | Preconditions | Test Steps | Expected Result |

|-----|-----|-----|-----|-----|-----|

| AT-01 | Acceptance | Customer Experience | Customer has an active account, valid payment method. | 1. The customer places an order and tracks the order status in real-time. 2. The customer receives push notifications for order updates. | The system provides a seamless customer experience, including real-time order tracking and push notifications. |

| AT-02 | Acceptance | Restaurant Experience | Restaurant has an active account, valid menu items. | 1. The restaurant manages their menu items and receives payment notifications. 2. The system updates the restaurant's menu and order status. | The system provides a seamless restaurant experience, including menu management and payment notifications. |

| AT-03 | Acceptance | Delivery Partner Experience | Delivery partner has an active account, valid delivery location. | 1. The delivery partner receives new delivery requests and tracks their progress in real-time. 2. The system updates the delivery partner's location and notifies the customer of the delivery update. | The system provides a seamless delivery partner experience, including real-time tracking and notifications. |

Please note that these test cases are just examples and may need to be modified or expanded based on the specific requirements of the project.